RetinaCare GUI manual (auto-extracted 2026-10-31)

Source of truth is the code; each row is file:line:text.

Pages and navigation (29 hits)

Where each page is declared and routed.

```
main.rs:160: const PAGES: [AppPage; 11] = [
main.rs:226: let active = PAGES
main.rs:255: if PAGES[i] == AppPage::PreviousChats {
main.rs:258: if PAGES[i] == AppPage::Patients {
main.rs:263: self.active_page = PAGES[i];
main.rs:2756: AppPage::Home => {
main.rs:2782: AppPage::Report | AppPage::ReportDoc | AppPage::Assessment => {
main.rs:2814: AppPage::Patients => {
main.rs:2840: AppPage::PreviousChats => {
main.rs:2874: AppPage::Timeline => {
main.rs:2903: AppPage::Explainability => {
main.rs:2918: AppPage::LesionAnalysis => {
main.rs:2930: AppPage::VesselAnalysis => {
main.rs:2953: AppPage::ResearchFindings => {
main.rs:2973: AppPage::Plugins => {
main.rs:3076: AppPage::Home => {
main.rs:3755: AppPage::Assessment => {
main.rs:3780: AppPage::ReportDoc => {
main.rs:3803: AppPage::Report => {
main.rs:4010: AppPage::PreviousChats => {
main.rs:4051: AppPage::Timeline => {
main.rs:4078: AppPage::Patients => {
main.rs:4087: AppPage::Explainability => {
main.rs:4090: AppPage::LesionAnalysis | AppPage::VesselAnalysis |
AppPage::ResearchFindings => {
main.rs:4094: AppPage::Plugins => {
science_app.rs:455: AppPage::LesionAnalysis => render_lesion_page(ui, rect, &mut
state, &inputs),
science_app.rs:456: AppPage::VesselAnalysis => render_vessel_page(ui, rect, &mut
state, &inputs),
engine/pdf.rs:7:const MAX_PAGES: usize = 100;
engine/pdf.rs:49: for &page_num in page_nums.iter().take(MAX_PAGES) {
```

Buttons and clickable labels (24 hits)

Every button label with its source line; the handler sits within a few lines of it.

```
ui2/composer.rs:387: if menu_item(ui, p, &format!("Remove {}", a.display_name)) {
ui2/composer.rs:473: if menu_item(ui, p, "Add image (PNG, JPG)") {
ui2/composer.rs:477: if menu_item(ui, p, "Add PDF report") {
ui2/chats_page.rs:210: if ui.button("Delete everything").clicked() {
ui2/chats_page.rs:214: if ui.button("Cancel").clicked() {
main.rs:2051: if ui.button("Open in browser").clicked() {
main.rs:2132: if ui.button("Open in browser").clicked() {
main.rs:2135: if ui.button("Open in app").clicked() {
main.rs:3680: if ui.button("Close").clicked() {
cloud_app.rs:141: if ui.add_enabled(ready && !busy, egui::Button::new("Sign
in")).clicked() || (enter && ready && !busy) {
ui/history.rs:399: .button(egui::RichText::new("Reset
Filters").size(12.0).color(theme::ACCENT_BLUE()))
ui/timeline.rs:243: .button(egui::RichText::new("Reset
Filters").size(12.0).color(theme::ACCENT_BLUE()))
ui/chat_cards.rs:167: Button::new(RichText::new("Assign to
patient...").size(11.5).color(theme::DIR_TEXT()))
ui/chat_cards.rs:185: Button::new(RichText::new("Save to
examination...").size(11.5).color(theme::DIR_TEXT()))
ui/chat_cards.rs:242: let open = Button::new(RichText::new("Open
report").size(11.5).color(theme::DIR_TEXT()))
ui/save_exam.rs:196: if ui.button("Cancel").clicked() {
ui/explain.rs:965: if ui.button("Cancel").clicked() {
ui/explain.rs:1128: if ui.button("Fit").clicked() {
ui/explain.rs:1339: .add_enabled(can, egui::Button::new("Export overlay PNG"))
ui/explain.rs:1345: .add_enabled(can, egui::Button::new("Export JSON"))
ui/explain.rs:1511: egui::Button::new("View"),
ui/explain.rs:1522: egui::Button::new("Compare"),
```

```
ui/explain.rs:1712: if ui.button("Open Report page (export PDF)").clicked() {
plugins/mod.rs:188: Button::new(RichText::new("All").size(12.0).color(if is_all {
```

Animations (19 hits)

Every animation primitive: bool/value tweens, eased curves, timed repaints.

```
ui2/attach_stack.rs:209: let t = ctx.animate_bool_with_time(key.with("t"), was_open
|| resp.hovered(), 0.28);
ui2/shell.rs:53:pub fn smootherstep(t: f32) -> f32 {
ui2/shell.rs:244: let t = ctx.animate_bool_with_time_and_easing(
ui2/shell.rs:248: smootherstep,
ui2/composer.rs:206: let t = ctx.animate_bool_with_time_and_easing(
ui2/chats_page.rs:4:use super::shell::smootherstep;
ui2/chats_page.rs:293: let t = ui.ctx().animate_bool_with_time_and_easing(
ui2/chats_page.rs:297: smootherstep,
ui2/widgets.rs:172: .animate_bool_with_time_and_easing(resp.id, *value, 0.2,
gentle);
ui2/console.rs:94: let t =
gentle(ctx.animate_bool_with_time(Id::new("ui2_console_anim"), self.open, 0.35));
main.rs:933: crate::engine::ollama::release_vram();
main.rs:2181: ctx.request_repaint_after(std::time::Duration::from_millis(500));
main.rs:2661: let sidebar_t = ui.ctx().animate_bool_with_time_and_easing(
main.rs:3078: let dock_t = ui.ctx().animate_bool_with_time_and_easing(
main.rs:5517: // Verify egui context registers request_repaint_after without panic
main.rs:5519: ctx.request_repaint_after(std::time::Duration::from_millis(500));
explain_app.rs:509:
ctx.request_repaint_after(std::time::Duration::from_millis(250));
engine/ollama.rs:100:pub fn release_vram() {
ui/web_cards.rs:112: .request_repaint_after(std::time::Duration::from_millis(50));
```

Pipeline stages (51 hits)

Screening, masks, grading and explainability entry points.

```
main.rs:442: analysis_stage: AnalysisStage::Idle,
main.rs:1158: == cascade::StageKey::from_stage(cascade::AnalysisStage::Segmentation(
domain.rs:13: AnalysisStage::Idle => "Idle",
domain.rs:14: AnalysisStage::ReportReady => "Report Ready",
domain.rs:217: stage: Some(AnalysisStage::ReportReady),
cascade_flow_tests.rs:223: stage: AnalysisStage::Screening,
cascade_flow_tests.rs:275: AnalysisStage::Segmentation(Stage2ModelKind::Ma),
cascade_flow_tests.rs:276: AnalysisStage::Segmentation(Stage2ModelKind::He),
cascade_flow_tests.rs:277: AnalysisStage::Segmentation(Stage2ModelKind::Ex),
cascade_flow_tests.rs:278: AnalysisStage::Segmentation(Stage2ModelKind::Cws),
cascade_flow_tests.rs:279: AnalysisStage::Segmentation(Stage2ModelKind::Vessel),
cascade_flow_tests.rs:280: AnalysisStage::Grading,
cascade_flow_tests.rs:454: stage: AnalysisStage::Screening,
cascade_flow_tests.rs:474: stage: AnalysisStage::Screening,
engine/cascade.rs:61: StageKey::Screening,
engine/cascade.rs:62: StageKey::Ma,
engine/cascade.rs:63: StageKey::He,
engine/cascade.rs:64: StageKey::Ex,
engine/cascade.rs:65: StageKey::Cws,
engine/cascade.rs:66: StageKey::Vessel,
engine/cascade.rs:67: StageKey::Grading,
engine/cascade.rs:84: AnalysisStage::Screening => Self::Screening,
engine/cascade.rs:85: AnalysisStage::Segmentation(Stage2ModelKind::Ma) => Self::Ma,
engine/cascade.rs:86: AnalysisStage::Segmentation(Stage2ModelKind::He) => Self::He,
engine/cascade.rs:87: AnalysisStage::Segmentation(Stage2ModelKind::Ex) => Self::Ex,
engine/cascade.rs:88: AnalysisStage::Segmentation(Stage2ModelKind::Cws) =>
Self::Cws,
engine/cascade.rs:89: AnalysisStage::Segmentation(Stage2ModelKind::Vessel) =>
Self::Vessel,
engine/cascade.rs:90: AnalysisStage::Grading => Self::Grading,
engine/cascade.rs:290: self.stages = StageKey::ALL
engine/cascade.rs:317: if let Some(row) = self.row(StageKey::from_stage(*stage)) {
engine/cascade.rs:323: if let Some(row) = self.row(StageKey::from_stage(*stage)) {
engine/cascade.rs:330: if let Some(row) = self.row(StageKey::from_stage(*stage)) {
engine/cascade.rs:337: if let Some(row) = self.row(StageKey::Grading) {
engine/cascade.rs:682: key:
StageKey::from_stage(AnalysisStage::Segmentation(m.model_kind)),
engine/cascade.rs:708: .map(|f| StageKey::from_stage(f.stage).label().to_string()),
engine/cascade.rs:716: StageKey::from_stage(t.stage).label().to_string(),
engine/cascade.rs:1162: stage: AnalysisStage::Screening
```

```
engine/cascade.rs:1274: stage: AnalysisStage::Grading,
engine/cascade.rs:1329: stage: AnalysisStage::Screening,
ui/viewer.rs:999: "run_stage1_screening_btn",
ui/cascade_cards.rs:60: StageKey::Screening => "Screening",
ui/cascade_cards.rs:61: StageKey::Ma => "MA",
ui/cascade_cards.rs:62: StageKey::He => "HE",
ui/cascade_cards.rs:63: StageKey::Ex => "EX",
ui/cascade_cards.rs:64: StageKey::Cws => "CWS",
ui/cascade_cards.rs:65: StageKey::Vessel => "Vessel",
ui/cascade_cards.rs:66: StageKey::Grading => "Grading",
ui/cascade_cards.rs:562: key: StageKey::Screening,
ui/science.rs:649: if !matches!(m.key, crate::engine::cascade::StageKey::Vessel) {
ui/science.rs:895: .find(|m| m.key == crate::engine::cascade::StageKey::Vessel)
ui/science.rs:915: .find(|m| m.key == crate::engine::cascade::StageKey::Vessel)
```

Chat and AI (18 hits)

Prompt, grounding and guardrails.

```
engine/ollama.rs:23:const SYSTEM_PROMPT: &str = "You are the chat assistant inside
RetinaCare, a research prototype for retinal image analysis. \
engine/ollama.rs:161:pub fn build_messages(request: &ChatRequest) ->
Vec<serde_json::Value> {
engine/ollama.rs:162: let mut system = SYSTEM_PROMPT.to_string();
engine/ollama.rs:219:pub fn ungrounded_numbers(answer: &str, given: &str) ->
Vec<String> {
engine/ollama.rs:232:pub fn stage_status_reply(prompt: &str, results: &str) ->
Option<String> {
engine/ollama.rs:285:pub fn plain_dashes(text: &str) -> String {
engine/ollama.rs:393: .and_then(|r| stage_status_reply(&request.prompt, r))
engine/ollama.rs:529: text: plain_dashes(text),
engine/ollama.rs:569: let nums = ungrounded_numbers(&answer, &given);
engine/ollama.rs:650: assert!(ungrounded_numbers("Score 0.065 is under 0.2392.",
given).is_empty());
engine/ollama.rs:652: ungrounded_numbers("For Stage II, 234 MA, 12 HE.", given),
engine/ollama.rs:655: assert!(ungrounded_numbers("1. First point\n2. Second point",
given).is_empty());
engine/ollama.rs:673: assert_eq!(plain_dashes("a \u{2014} b \u{2013} c"), "a - b -
c");
engine/ollama.rs:675: assert!(SYSTEM_PROMPT.contains("never use an em dash"));
engine/ollama.rs:681: let a = stage_status_reply("why were stage 1 and 3 did not
run", all).unwrap();
engine/ollama.rs:683: assert!(stage_status_reply("what does grade 2 mean",
all).is_none());
engine/ollama.rs:685: let b = stage_status_reply("stage III did not run, why",
some).unwrap();
engine/ollama.rs:837: ungrounded_numbers(&a, &format!("{ctx}\n{q}")).is_empty(),
```

PDF report (8 hits)

Report generation.

```
main.rs:1727: fn build_pdf_document(&self) -> retinacare_reports::PdfDocument {
engine/report.rs:200:pub fn build_report_document(ctx: &ReportContext) ->
retinacare_reports::PdfDocument {
engine/report.rs:355: title: "Retinal Fundus Examination Report".to_string(),
ui/viewer.rs:2264:fn render_clinical_report_overview(
ui/viewer.rs:2859:fn render_report_actions_footer(
ui/viewer.rs:2982:fn render_pdf_export_modal(
ui/viewer.rs:3489:fn render_report_subroute_toolbar(
ui/viewer.rs:3564:fn render_pdf_export_screen(
```

Storage and security (44 hits)

Encrypted store, hashing, wipes.

```
main.rs:33:use security::{compute_sha256, scrub_image_metadata,
validate_attachment_path};
main.rs:765: let sha256_hash = Some(compute_sha256(&file_bytes));
main.rs:987: let hash =
crate::security::compute_sha256(&std::fs::read(&path).ok()?);
main.rs:998: .is_ok_and(|b| crate::security::compute_sha256(&b) == hash);
cloud_app.rs:8:use zeroize::Zeroize;
```

```
cloud_app.rs:142: let password = zeroize::Zeroizing::new(std::mem::take(&mut
self.cloud_password));
cloud_app.rs:143: self.cloud_password.zeroize();
cloud_app.rs:156: self.cloud_password.zeroize();
chat_app.rs:150: (Some(want), Ok(bytes)) if crate::security::compute_sha256(&bytes)
!= *want => {
chat_app.rs:455: pub(crate) fn delete_all_chats(&mut self, ctx: &egui::Context) {
chat_app.rs:1207: sha256_hash: Some(crate::security::compute_sha256(b"%PDF-1.4
test")),
science_app.rs:9:use crate::security::compute_sha256;
science_app.rs:104: if compute_sha256(&bytes) != job.image_sha256 {
security.rs:70:pub struct ZeroizedBuffer {
security.rs:74:impl ZeroizedBuffer {
security.rs:92:impl Drop for ZeroizedBuffer {
security.rs:103:pub fn compute_sha256(data: &[u8]) -> String {
security.rs:123: let actual = compute_sha256(data);
security.rs:169:/// NIST SP 800-38D AES-256-GCM Authenticated Decryption returning
zeroized memory buffer
security.rs:173:) -> Result<ZeroizedBuffer, SecurityError> {
security.rs:191: Ok(ZeroizedBuffer::new(decrypted))
security.rs:367: let empty_hash = compute_sha256(b"");
security.rs:372: let retina_hash = compute_sha256(b"RetinaCare AI Screening");
security.rs:414: let buffer = ZeroizedBuffer::new(vec![0xAA, 0xBB, 0xCC, 0xDD]);
engine/supabase.rs:15:use zeroize::Zeroizing;
engine/keystore.rs:6:use zeroize::Zeroizing;
engine/papers.rs:153: .map(|b| crate::security::compute_sha256(&b)[..12].to_owned())
engine/patient.rs:69: /// Connected to live, unlocked SQLCipher backend.
engine/patient.rs:71: /// Backend SQLCipher store exists but requires password
unlock.
engine/patient.rs:93: write!(f, "SQLCipher database is locked. Master password
required.")
engine/patient.rs:264:// Live Patient Service: encrypted SQLCipher store
(retinacare-persistence)
engine/patient.rs:268:/// frame without querying SQLCipher every frame.
engine/cloud.rs:11:use zeroize::Zeroizing;
engine/store.rs:65: let dest = dir.join(format!("{}.{ext}",
crate::security::compute_sha256(&bytes)));
engine/store.rs:245: let password = zeroize::Zeroizing::new(password);
ui/patients.rs:63: pub unlock_password: zeroize::Zeroizing<String>,
ui/patients.rs:64: pub unlock_confirm: zeroize::Zeroizing<String>,
ui/patients.rs:93: unlock_password: zeroize::Zeroizing::new(String::new()),
ui/patients.rs:94: unlock_confirm: zeroize::Zeroizing::new(String::new()),
ui/patients.rs:352: "● SQLCipher Connected",
ui/patients.rs:357: "■ SQLCipher Locked (Auth Required)",
ui/patients.rs:612: "Choose a password of 8 or more characters. Records are
encrypted with SQLCipher. The password cannot be recovered if lost."
ui/patients.rs:686: zeroize::Zeroize::zeroize(&mut *state.unlock_confirm);
ui/patients.rs:2105: "Notice: SQLCipher store is currently locked. Action requires
live credentials.",
```

Tests (286 hits)

Every unit test marker (count shows coverage).

```
ui2/attach_stack.rs:329: #[test]
ui2/shell.rs:501: #[test]
ui2/shell.rs:508: #[test]
ui2/shell.rs:516: #[test]
ui2/composer.rs:588: #[test]
ui2/composer.rs:594: #[test]
ui2/chats_page.rs:479: #[test]
ui2/chats_page.rs:490: #[test]
ui2/widgets.rs:291: #[test]
ui2/widgets.rs:309: #[test]
ui2/widgets.rs:321: #[test]
ui2/console.rs:207: #[test]
ui2/svg.rs:143: #[test]
ui2/svg.rs:152: #[test]
ui2/theme.rs:135: #[test]
ui2/theme.rs:157: #[test]
ui2/theme.rs:169: #[test]
ui2/theme.rs:204: #[test]
main.rs:4554: #[test]
main.rs:4574: #[test]
main.rs:4584: #[test]
main.rs:4590: #[test]
```

```
main.rs:4599: #[test]
main.rs:4614: #[test]
main.rs:4627: #[test]
main.rs:4638: #[test]
main.rs:4646: #[test]
main.rs:4657: #[test]
main.rs:4672: #[test]
main.rs:4692: #[test]
main.rs:4701: #[test]
main.rs:4710: #[test]
main.rs:4729: #[test]
main.rs:4738: #[test]
main.rs:4764: #[test]
main.rs:4781: #[test]
main.rs:4851: #[test]
main.rs:4869: #[test]
main.rs:4890: #[test]
main.rs:4909: #[test]
main.rs:4924: #[test]
main.rs:4938: #[test]
main.rs:4947: #[test]
main.rs:4961: #[test]
main.rs:4969: #[test]
main.rs:5002: #[test]
main.rs:5031: #[test]
main.rs:5059: #[test]
main.rs:5072: #[test]
main.rs:5089: #[test]
main.rs:5129: #[test]
main.rs:5173: #[test]
main.rs:5203: #[test]
main.rs:5225: #[test]
main.rs:5234: #[test]
main.rs:5271: #[test]
main.rs:5300: #[test]
main.rs:5328: #[test]
main.rs:5369: #[test]
main.rs:5433: #[test]
main.rs:5489: #[test]
main.rs:5503: #[test]
main.rs:5523: #[test]
main.rs:5554: #[test]
main.rs:5577: #[test]
main.rs:5596: #[test]
main.rs:5637: #[test]
main.rs:5661: #[test]
chat_app.rs:1193: #[test]
chat_app.rs:1239: #[test]
chat_app.rs:1276: #[test]
chat_app.rs:1283: #[test]
chat_app.rs:1315: #[test]
chat_app.rs:1351: #[test]
chat_app.rs:1369: #[test]
chat_app.rs:1387: #[test]
chat_app.rs:1417: #[test]
domain.rs:942: #[test]
domain.rs:950: #[test]
domain.rs:956: #[test]
domain.rs:962: #[test]
domain.rs:968: #[test]
domain.rs:978: #[test]
domain.rs:1000: #[test]
domain.rs:1014: #[test]
domain.rs:1031: #[test]
domain.rs:1053: #[test]
domain.rs:1061: #[test]
domain.rs:1077: #[test]
domain.rs:1089: #[test]
domain.rs:1099: #[test]
domain.rs:1107: #[test]
domain.rs:1119: #[test]
domain.rs:1128: #[test]
domain.rs:1149: #[test]
domain.rs:1166: #[test]
domain.rs:1186: #[test]
security.rs:343: #[test]
security.rs:364: #[test]
```

```
security.rs:377: #[test]
security.rs:387: #[test]
security.rs:398: #[test]
security.rs:412: #[test]
security.rs:419: #[test]
security.rs:433: #[test]
security.rs:440: #[test]
security.rs:449: #[test]
security.rs:458: #[test]
security.rs:469: #[test]
cascade_flow_tests.rs:169:#[test]
cascade_flow_tests.rs:190:#[test]
cascade_flow_tests.rs:198:#[test]
cascade_flow_tests.rs:215:#[test]
cascade_flow_tests.rs:269:#[test]
cascade_flow_tests.rs:307:#[test]
cascade_flow_tests.rs:326:#[test]
cascade_flow_tests.rs:345:#[test]
cascade_flow_tests.rs:361:#[test]
cascade_flow_tests.rs:384:#[test]
cascade_flow_tests.rs:410:#[test]
cascade_flow_tests.rs:468:#[test]
cascade_flow_tests.rs:515:#[test]
cascade_flow_tests.rs:548:#[test]
cascade_flow_tests.rs:597:#[test]
cascade_flow_tests.rs:684:#[test]
cascade_flow_tests.rs:709:#[test]
cascade_flow_tests.rs:763:#[test]
cascade_flow_tests.rs:786:#[test]
cascade_flow_tests.rs:865:#[test]
cascade_flow_tests.rs:923:#[test]
cascade_flow_tests.rs:941:#[test]
cascade_flow_tests.rs:960:#[test]
cascade_flow_tests.rs:1013:#[test]
cascade_flow_tests.rs:1038:#[test]
cascade_flow_tests.rs:1062:#[test]
cascade_flow_tests.rs:1105:#[test]
cascade_flow_tests.rs:1171:#[test]
cascade_flow_tests.rs:1244:#[test]
engine/ollama.rs:621: #[test]
engine/ollama.rs:630: #[test]
engine/ollama.rs:647: #[test]
engine/ollama.rs:671: #[test]
engine/ollama.rs:678: #[test]
engine/ollama.rs:689: #[test]
engine/ollama.rs:698: #[test]
engine/ollama.rs:708: #[test]
engine/ollama.rs:719: #[test]
engine/ollama.rs:775: #[test]
engine/ollama.rs:817: #[test]
engine/ollama.rs:860: #[test]
engine/ollama.rs:878: #[test]
engine/chat.rs:114: #[test]
engine/chat.rs:150: #[test]
engine/supabase.rs:561: #[test]
engine/supabase.rs:579: #[test]
engine/supabase.rs:613: #[test]
engine/supabase.rs:663: #[test]
engine/supabase.rs:685: #[test]
engine/supabase.rs:704: #[test]
engine/supabase.rs:728: #[test]
engine/supabase.rs:757: #[test]
engine/keystore.rs:63: #[test]
engine/export.rs:630: #[test]
engine/export.rs:668: #[test]
engine/export.rs:696: #[test]
engine/export.rs:743: #[test]
engine/export.rs:777: #[test]
engine/export.rs:820: #[test]
engine/manifest.rs:103: #[test]
engine/cascade.rs:1080: #[test]
engine/cascade.rs:1094: #[test]
engine/cascade.rs:1124: #[test]
engine/cascade.rs:1151: #[test]
engine/cascade.rs:1175: #[test]
engine/cascade.rs:1221: #[test]
engine/cascade.rs:1292: #[test]
```

```
engine/cascade.rs:1316: #[test]
engine/cascade.rs:1342: #[test]
engine/cascade.rs:1358: #[test]
engine/cascade.rs:1373: #[test]
engine/cascade.rs:1584: #[test]
engine/cascade.rs:1604: #[test]
engine/gpu.rs:74: #[test]
engine/papers.rs:220: #[test]
engine/papers.rs:227: #[test]
engine/papers.rs:234: #[test]
engine/papers.rs:242: #[test]
engine/papers.rs:262: #[test]
engine/papers.rs:278: #[test]
engine/pdf.rs:81: #[test]
engine/patient.rs:967: #[test]
engine/patient.rs:992: #[test]
engine/patient.rs:1004: #[test]
engine/patient.rs:1026: #[test]
engine/patient.rs:1042: #[test]
engine/patient.rs:1052: #[test]
engine/patient.rs:1094: #[test]
engine/patient.rs:1160: #[test]
engine/patient.rs:1212: #[test]
engine/explain.rs:449: #[test]
engine/explain.rs:476: #[test]
engine/explain.rs:509: #[test]
engine/explain.rs:525: #[test]
engine/explain.rs:553: #[test]
engine/report.rs:376: #[test]
engine/report.rs:407: #[test]
engine/report.rs:442: #[test]
engine/report.rs:450: #[test]
engine/report.rs:527: #[test]
engine/report.rs:542: #[test]
engine/report.rs:552: #[test]
engine/report.rs:580: #[test]
engine/report.rs:593: #[test]
engine/report.rs:615: #[test]
engine/report.rs:642: #[test]
engine/cloud.rs:315: #[test]
engine/cloud.rs:374: #[test]
engine/cloud.rs:457: #[test]
engine/cloud.rs:516: #[test]
engine/store.rs:359: #[test]
engine/store.rs:403: #[test]
engine/store.rs:420: #[test]
engine/store.rs:430: #[test]
engine/store.rs:441: #[test]
engine/store.rs:470: #[test]
engine/store.rs:488: #[test]
engine/web.rs:203: #[test]
engine/web.rs:211: #[test]
engine/web.rs:216: #[test]
engine/web.rs:224: #[test]
engine/science.rs:306: #[test]
engine/science.rs:316: #[test]
engine/science.rs:336: #[test]
engine/science.rs:352: #[test]
engine/science.rs:366: #[test]
engine/science.rs:406: #[test]
engine/science.rs:414: #[test]
engine/backend.rs:687: #[test]
engine/backend.rs:709: #[test]
engine/backend.rs:718: #[test]
engine/backend.rs:732: #[test]
engine/backend.rs:746: #[test]
engine/backend.rs:765: #[test]
engine/backend.rs:773: #[test]
ui/viewer.rs:4075: #[test]
ui/viewer.rs:4092: #[test]
ui/viewer.rs:4105: #[test]
ui/viewer.rs:4125: #[test]
ui/viewer.rs:4143: #[test]
ui/composer.rs:474: #[test]
ui/composer.rs:486: #[test]
ui/composer.rs:495: #[test]
ui/composer.rs:513: #[test]
```

```
ui/cascade_cards.rs:506: #[test]
ui/cascade_cards.rs:517: #[test]
ui/cascade_cards.rs:532: #[test]
ui/cascade_cards.rs:554: #[test]
ui/save_exam.rs:222: #[test]
ui/save_exam.rs:230: #[test]
ui/explain.rs:1756: #[test]
ui/explain.rs:1775: #[test]
ui/explain.rs:1793: #[test]
ui/explain.rs:1808: #[test]
ui/explain.rs:1835: #[test]
ui/explain.rs:1851: #[test]
ui/explain.rs:1873: #[test]
ui/explain.rs:1883: #[test]
ui/explain.rs:1891: #[test]
ui/explain.rs:1912: #[test]
ui/explain.rs:1983: #[test]
ui/components.rs:125: #[test]
ui/science.rs:1552: #[test]
ui/science.rs:1561: #[test]
ui/science.rs:1575: #[test]
ui/science.rs:1582: #[test]
ui/science.rs:1592: #[test]
ui/markdown.rs:531: #[test]
ui/markdown.rs:541: #[test]
ui/markdown.rs:549: #[test]
plugins/pipeline.rs:186: #[test]
plugins/pipeline.rs:194: #[test]
plugins/pipeline.rs:208: #[test]
plugins/icons.rs:130: #[test]
plugins/icons.rs:160: #[test]
plugins/mod.rs:741: #[test]
plugins/mod.rs:748: #[test]
```